

Formula Student Brake Pedal Assembly

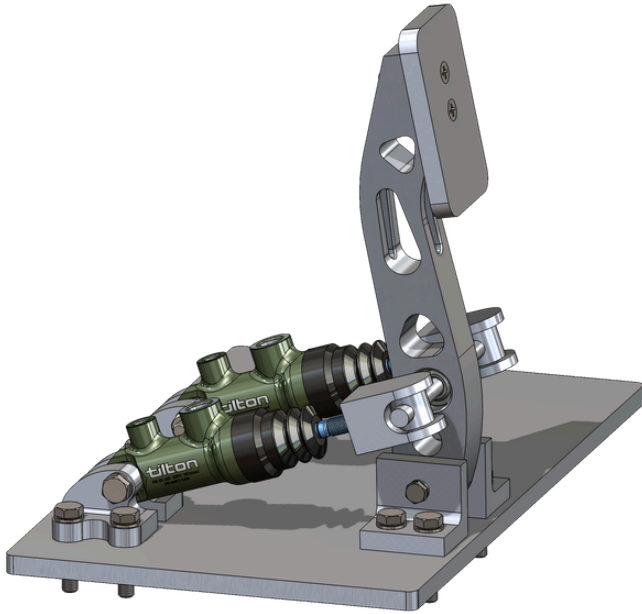
Designed and optimised a Formula Student brake pedal assembly in Al 7075-T6 for CNC manufacture

Achieved:

- **30.5%** mass reduction
- **1.97** safety factor
- Deformation under **2mm** at peak load

How:

- Designed pedal body and bracket in SolidWorks to a 4:1 pedal ratio PDS.
- Validated with hand calculations and multi-load-case FEA in ANSYS Static Structural.
- Applied topology optimisation and redesigned for 3-axis CNC manufacturability.



Software and Techniques:

SolidWorks, ANSYS Mechanical, Structural Optimisation, DFM

Formula Student Intake Assembly

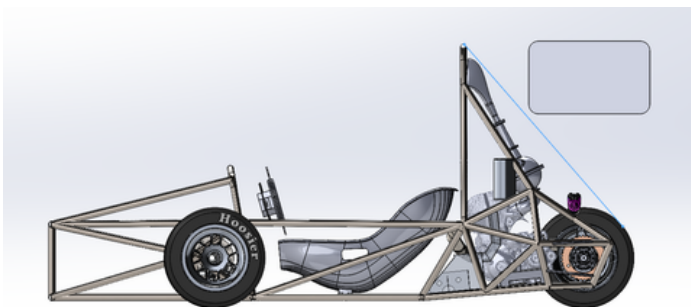
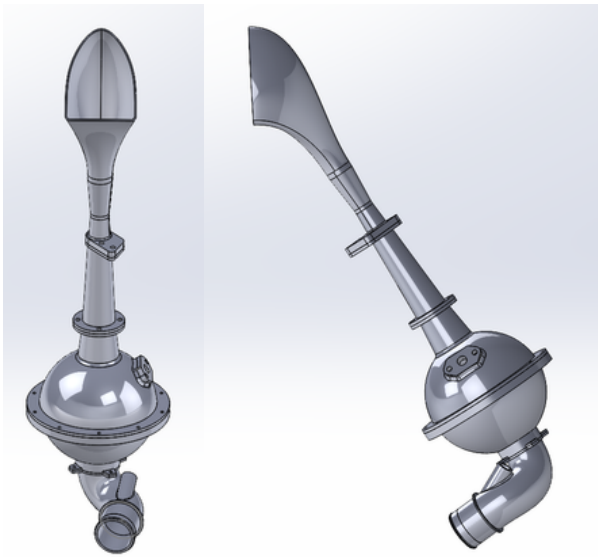
Designed and fabricated a Formula Student Intake Assembly in Carbon Fibre Reinforced Nylon.

Achieved:

- **Acoustic tuning:** Tuned runner geometry for peak power delivery at 7000 RPM
- **Packaging Compliance:** Assembly routed to strictly adhere to chassis and firewall constraints
- **Fabrication:** Delivered a complete, multi-piece 3D-printed assembly optimized for rapid prototyping.

How:

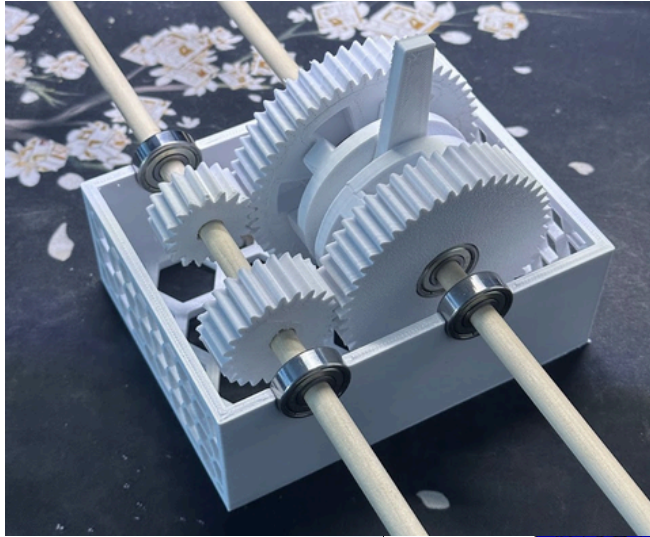
- Utilized Helmholtz resonance equations to determine optimal runner length and volume
- Modeled plenum and runner geometry in SolidWorks to integrate with chassis constraints
- Sliced and fabricated the multi-section geometry using FDM 3D printing



Software and Techniques:

SolidWorks, ANSYS Mechanical & Fluent, DFM, FDM 3D-Printing, Packaging Optimisation

Multi-Speed Manual Transmission Gearbox



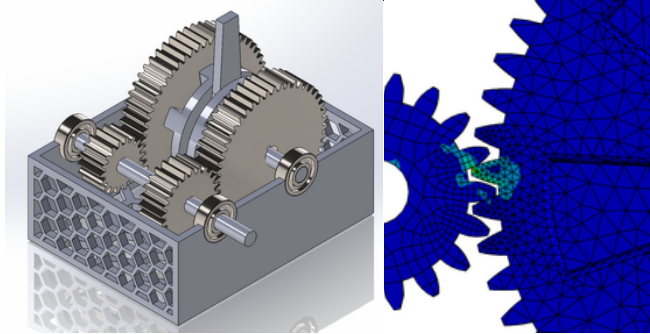
Designed, simulated and prototyped a multi-speed manual transmission gearbox, optimised for Additive Manufacturing.

Achieved:

- **3:1** (1st gear) and **2:1** (2nd gear) reduction ratios
- **2.05** safety factor on critical components
- **100% functional prototype**

How:

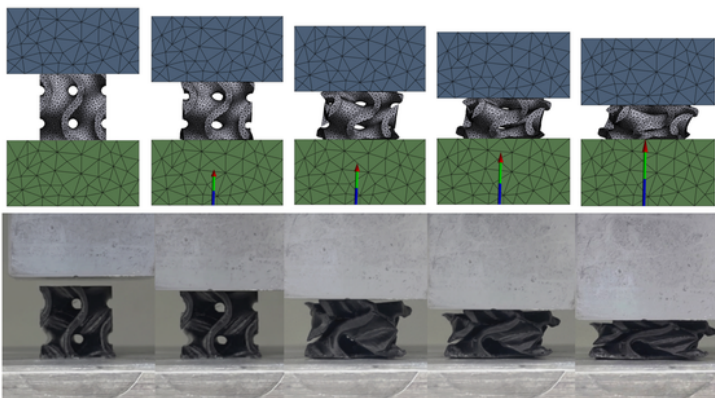
- Designed gearbox assembly in SolidWorks after calculating gear ratios and predicted loads.
- Validated with FEA in ANSYS Static Structural.
- Additively manufactured and assembled gearbox prototype for gear ratio testing and functionality.



Software and Techniques:

SolidWorks, ANSYS Mechanical, Gear Design, Rapid Prototyping (FDM)

Bio-Inspired Crash Structure - Final Year Project



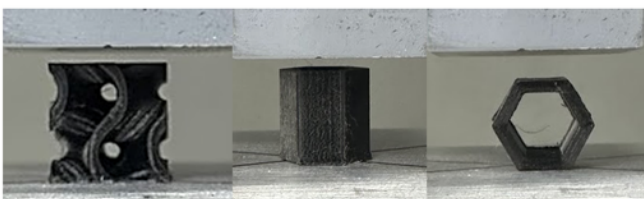
Designed, simulated and experimentally validated a bio-inspired gyroid TPMS crash structure for formula student applications.

Achieved:

- Gyroid TPMS achieved CFE of **0.893**
- FEA validated against physical testing with **9%** CFE error
- **9** specimen designed, 3D printed and physically tested.
- **First Class Grade.**

How:

- Designed and 3D printed PETG gyroid and hexagonal prism specimens for quasi-static compression testing.
- Validated deformation behaviour using ANSYS Explicit Dynamics FEA.
- Compared energy absorption metrics (SEA, CFE, peak force) across all geometries and orientations.



Software and Techniques:

SolidWorks, nTop, ANSYS Explicit Dynamics, FDM 3D-Printing, Quasi-Static compression testing.